

TLDR: Senior AI Engineer, 5+ years of experience driving cloud-based AI/ML products to market in the AI for Science, Self Driving, and Enterprise Data spaces. **Skills:** Python, Rust, TypeScript, PyTorch, Tensorflow, LangChain, Langfuse, LlamaIndex, AWS, GCP, Docker, SQL, REST, websockets, French, Spanish.

Senior AI Engineer (Nov 2025 - Present) | AI Engineer (Apr 2024 - Nov 2025) — [Materiom](#)

Funded by Google, end-to-end development of production-ready AI software for bio-based materials, focusing on generative AI, LLMs, integrating tools and knowledge bases from materials science and biochemistry:

- **Materiom AI:** Agentic RAG application with access to 40000+ materials and various search APIs, serving 10000+ user sessions since its deployment, and 10+ enterprise partnerships.
- **Materiom Commons:** web-based platform accelerating bio-based material innovation, catering to 25000+ registered users across 130 countries with 99.99% uptime.
- **Neural models for material synthesis and material property prediction** trained on experimental data from partner scientific labs across the world. Trialed with multiple enterprise partners.
- **Evaluation pipelines** for automated, continuous model monitoring in production, including expert-led metrics to enable rapid diagnostics and model improvement.
- **Engaged in key investor and partner events** to expand our customer base and establish strategic partnerships in Europe, the US, and the UK.

AI Engineer (Sep 2021 - Apr 2024) — [dRISK](#) (absorbed by [Nexar Inc.](#))

Led the design, development, and deployment of cloud-based AI-human collaboration features at Conode (previously dRISK):

- **Enterprise Data Solution:** web app built with TypeScript, Rust, and Python, enabling non-technical decision-makers to navigate and understand large datasets graphically.
- **AV Perception Retraining Platform:** Allows users to graphically identify and fix perception failures through data curation, model retraining, fine-tuning, and performance tracking.
- **LLM-Powered Agent:** Enables non-technical users to interact with data exploration tools via natural language, hosted on a Python Flask server with websocket communication.
- **Led multiple customer-facing projects from PoC to mature relationships** with major clients like NVIDIA, Luminar, TATA Motors, SAE and others.
- **Engaged in key investor events** to expand our customer base and form strategic partnerships, especially in SE Asia and the UK.

AI Researcher / Master's Thesis (Jun 2021 - Sep 2021) — [Neurolabs](#) & The University of Edinburgh

Thesis: *Graph Convolutional Neural Networks for Synthetic Data Generation using Unity* (Awarded Distinction). Researched and developed geometric deep learning pipelines to generate high-fidelity synthetic spatial data. Read more [here](#).

Education & Scholarships

MSc in Artificial Intelligence ([The University of Edinburgh](#), grade: distinction), MSc in Mechatronic Engineering ([Arts et Métiers ParisTech](#), grade: 17/20), BSc in Product Design Engineering ([EAFIT University](#), grade: 4.75/5.0). [Colfuturo Excellence Scholarship](#), [Eiffel Excellence Scholarship](#).

Patents

([Application No. PCT/US2024/032276](#)): ML algorithm using graph theory and A* search to augment trajectory data for training autonomous vehicle RL models.

Talks

AI for Materials Science, University of the Arts Central Saint Martins, [A day in the life of an AI Startup](#), EAFIT University, [Synthetic Data Generation for ML](#), Machine Learning Colombia